
Loss Landscape Geometry of Partial Differential Equation Emulators: Or, Gradient Alignment as a Mechanism for Symmetry Learning

Anonymous Author(s)

Affiliation

Address

email

Abstract

1 Neural emulators of solution operators to partial differential equations capable
2 of reliable generalization stand to accelerate scientific discovery by extrapolating
3 beyond training regimes and performing over varied initial condition classes. As-
4 sessing model generalization capabilities requires distinguishing solution operator-
5 consistent behavior from memorization of feature derived correlations, which can
6 be achieved by using influence functions to characterize how information is shared
7 across examples with different temporal indices and/or of different initial conditions.
8 We compute the hat-matrix metric-weighted gradient alignment by inverting the
9 relevant Hessian restricted to the mini-batch gradient-spanned subspace, enabling
10 exact measurement of cross-time and cross-class coherence during training. This
11 diagnostic is architecture-agnostic and can be efficiently computed. Our results
12 indicate that gradient alignment between training examples is governed by their
13 feature-space similarity. We find that gradient coherence across initial condition
14 classes is suppressed due to feature-space dissimilarity. Moreover, we quantify the
15 temporal decay of gradient alignment induced by standard one-step autoregressive
16 training, revealing that gradient coherence is not extended in time, which limits
17 long-horizon extrapolation performance. Cross-time influence exhibits a negative
18 correlation with rollout error across horizons, indicating that the measured response
19 geometry contains information predictive of inference-time degradation.

20 1 Introduction

21 Deep learning emulators for partial differential equation (PDE) solvers routinely achieve impressive
22 in-distribution accuracy Brandstetter *et al.* (2023); Herde *et al.* (2024); Takamoto *et al.* (2024); Lippe
23 *et al.* (2023); Gupta and Brandstetter (2022); Ohana *et al.* (2025), yet they often fail to respect the
24 fundamental symmetries of the governing equations Akhound-Sadegh *et al.* (2023); Gregory *et al.*
25 (2024); Gruver *et al.* (2022). This limitation undermines their ability to extrapolate and generalize,
26 raising the question: are such models truly learning physics, or merely fitting correlations present
27 in the training data? Explaining this gap requires probing not just the outputs, but also the learning
28 process Fort *et al.* (2020); Zhao *et al.* (2024).

29 Symmetries of the Navier-Stokes equations, namely translations, rotations, reflections, scalings, and
30 Galilean boosts, organize the solution space into orbits whose members are physically equivalent
31 Brandstetter *et al.* (2022). A model that has learned the solution operator will propagate information
32 seamlessly across these orbits: gradients of the loss with respect to parameters, evaluated on symmetry-
33 related inputs, should align, on account of equivariance; without such coherence, the resulting loss
34 differentials do not constructively influence one another, rendering the orbit decoupled. Measuring

35 cross-influence offers a diagnostic beyond standard forward-pass equivariance checks, exposing the
36 degree to which gradient signals are physically consistent.

37 If influence across group actions presents only weakly, the model is memorizing localized patterns
38 rather than learning physical processes Arpit *et al.* (2017); He and Su (2020); Chatterjee (2020).
39 Conversely, marked gradient alignment signals that the network has learned to couple symmetry-
40 related states, consistent with the behavior of a true solution operator. Our symmetry-aware gradient
41 diagnostic therefore isolates the process by which a model learns to generalize across orbits,
42 providing a principled tool to assess how architectural choices, loss design, and inductive biases
43 promote, or hinder, robust generalization.

44 2 Contributions

45 Our work contributes to three core themes of contemporary machine learning research. First, it
46 advances interpretability methods by using influence functions to directly probe training dynamics,
47 going beyond an analysis of forward-pass behavior alone. Second, it informs generalization theory
48 by framing symmetry learning as a problem of basin selection in the loss landscape, governed by
49 orbit-wise gradient coherence. Third, within scientific machine learning, it provides a principled
50 diagnostic for assessing if neural emulators have genuinely learned symmetries of the underlying
51 solution operator.

52 We extend a previously developed gradient-based explainability framework Anonymous (2026) to
53 determine why data-driven PDE emulators often fail to learn and exploit physical symmetries. We
54 introduce a geometry-aware, symmetry-conditioned gradient-influence diagnostic that probes how
55 training updates propagate across symmetry group orbits, specifically the dihedral group of rotations
56 and reflections, in addition to specific discrete translations. Our diagnostic is architecture- and domain-
57 agnostic, provided gradients and group actions are defined. We pair this gradient alignment analysis
58 with forward pass equivariance error tests to produce a coherent audit of symmetry learning. This
59 study of how gradient coherence facilitates the flow of information through a group orbit supplements
60 prediction-based evaluation metrics by determining if the underlying learning dynamics that govern
61 generalization are symmetry-compatible. This analysis reveals that disrespect of symmetry may
62 present not only in representation space but also in the local geometry, which need not support a
63 coherent update structure across symmetry-related inputs.

64 In this work, generalization refers to the standard notion of test-risk on unseen data; our focus is
65 on equivariance consistency and its mechanistic underpinnings. The proposed orbit-wise gradient
66 coherence is a local property of the trained model’s loss landscape. Exact equivariance implies
67 uniform coherence, but the converse need not hold, for gradients can be coherent on their approach
68 to an equivariant converged state. We therefore use coherence as a diagnostic for determining if
69 training updates couple symmetry-related states, and we relate it empirically to forward equivariance
70 error across dihedral and translational transformations. Together, these components provide a concise
71 framework for evaluating symmetry-consistent behavior via both forward-pass consistency and probes
72 of the learning dynamics

73 3 Method

74 We compare a UNet (13M parameters, 4 down-sampling blocks, 24 embedding channels) and a Vision
75 Transformer (ViT; 5M parameters, 6 layers, 256 channels) trained as emulators for two-dimensional
76 compressible Euler flows from PDEGym Herde *et al.* (2024). For data, we selected three classes
77 of Riemann-type initial conditions (CE-RP, CE-RPUI, CE-CRP), each with 5,000 trajectories of 16
78 time steps. Each state snapshot is a 128×128 grid of mass density, Cartesian momentum density,
79 and energy density. Models were trained autoregressively to emulate the Euler evolution operator. To
80 complement the compressible Euler results, we also apply our analysis to models trained on velocity
81 fields generated by the Navier-Stokes (NS) equations using, NS-BB, NS-Gauss, and NS-Sines initial
82 conditions. These flows occupy a qualitatively different feature space, characterized by smoother,
83 viscosity-regularized dynamics and vorticity-dominated structure.

84 Optimization used Adam with learning rate 5×10^{-4} and weight decay $\lambda = 10^{-4}$ on mini-batches
85 of $N = 48$ transitions. The cost function was a scaled mean-squared error (SMSE), that normalizes
86 errors by channel RMS to balance large and small-amplitude features, ensuring shocks and wavefronts

are captured while retaining sensitivity to quiescent flows, in addition to rendering dimensionless the influence matrix of interest. Both models were trained in distributed mode on two 40GB A100 GPUs using Lux.jl Pal (2023a,b) and Zygote.jl Innes (2018), with three seeds controlling initialization and dataset splits. Results are reported with quantile range bars to capture variability across seeds and mini-batches.

To evaluate our models, we compute the influence function, which can be expressed as the Lie derivative of the cost along gradient flow induced by individual test examples. Let $V^\mu = -\chi^{\mu\nu} \partial_\nu C_x$ denote the vector field generated by the loss evaluated on an example x . The influence of this update on the loss evaluated at the transformed input gx is given by

$$\mathcal{L}_V C_{gx} = (\partial_\mu C_{gx}) \chi^{\mu\nu} (-\partial_\nu C_x), \quad (1)$$

where $\chi_{\mu\nu} = \eta_{\mu\nu} + \lambda \delta_{\mu\nu}$ is the regularized Gauss-Newton metric Jacot *et al.* (2020), and $\chi^{\mu\nu}$ denote the elements of χ^{-1} Absil *et al.* (2008). Einstein summation convention is implicit and we use standard index raising notation from differential geometry. Equation 1 recovers the familiar influence function definition as a metric-weighted overlap between gradients derived from the cost evaluated on an example x and the transformed counterpart gx Fort and Ganguli (2019). Intuitively, the influence function measures whether a gradient update induced by one example decreases (or increases) the loss of another. When applied to symmetry-related inputs, it quantifies whether learning signals propagate coherently along symmetry orbits.

In regression, the Gauss-Newton matrix plays the role of a Fisher-information analog by supplying the Jacobian-induced metric on parameter space Martens (2020). For each model seed, the influence function is evaluated across eighteen test mini-batches comprising three distinct initial conditions over sixteen time-steps. The resulting influence matrices are standardized Lu *et al.* (1997); H  ritier and Ronchetti (1994) by normalizing with respect to the empirical variance of perturbations in each mini-batch, so that unity defines the natural baseline associated with unstructured stochastic variability. Deviations from this scale therefore quantify influence beyond random training noise, enabling principled interpretation of both self- and cross-response structure.

For further discussion of data selection, model selection, an intuitive derivation of the influence function, and details on its computation, see Appendix A, Appendix B, Appendix C, and Appendix D, respectively.

4 Results

Our evaluation jointly considers equivariance error, which probes forward-pass consistency under symmetry transformations, and influence function matrix elements, which quantify the alignment of parameter updates induced by symmetry-related inputs. The influence function reveals whether learning dynamics propagate information coherently across symmetry-related states, exposing whether a model is genuinely learning symmetries or merely fitting correlations in the data. Convergence to basins that respect the symmetries of the underlying problem is both essential for generalization and a persistent challenge for current architectures. Throughout this section, we test the central hypothesis that forward-pass equivariance emerges precisely when training dynamics propagate gradient information coherently along symmetry orbits. We find that forward-pass equivariance error is characterized by the extent to which our models and training regimes support symmetry-coherent gradients.

We begin by analyzing symmetry learning with respect to the dihedral group D_4 , the set of discrete rotations and reflections admissible under the datasets used for training. Our models successfully generalize on those group transformations for which gradients are propagated coherently through the loss landscape. Where generalization fails, we show that training dynamics do not support full orbit-wise coupling and visualize how data-induced anisotropies are absorbed into the learned loss landscape geometry.

Next, we consider the translation group. In this case, our UNets exhibit constructive gradient coherence that is nearly uniform. On the other hand, our ViTs distribute influence non-uniformly but support larger influence on privileged group elements. These observations highlight a trade-off between inductive bias and optimization flexibility. Enforcing uniform coupling across all group elements can constrain the space of admissible update directions and, in some settings, impede optimization Zhang (2019); Azulay and Weiss (2019); Kayhan and van Gemert (2020) by inducing

a mean gradient direction that is inconsistent with individual update directions, otherwise known as type-II gradient misalignment Wang *et al.* (2025). In our translation group related experiments, we observe exclusively positive influence values, indicating that updates are not antagonistic across group elements. The absence of negative influence indicates that optimization does not suffer from conflicting updates across group elements; instead, the distinction lies in how learning signal is distributed: architectures with weaker inductive bias are free to concentrate influence on a subset of symmetry transformations, enabling rapid training loss reduction via a strong gradient response, but limiting generalization capabilities across the group. This contrast illustrates how architectural constraints can promote principled symmetry learning at the cost of optimization flexibility, while unconstrained models optimize efficiently yet may only partially learn the symmetry structure, motivating approximately constrained approaches that balance these effects Wang *et al.* (2022).

Convergence to loss basins that respect the symmetries of the underlying PDE is essential for robust generalization, yet remains a persistent challenge for current architectures. The following results demonstrate that influence functions provide a direct probe of this phenomenon by revealing whether learning dynamics couple physically equivalent states.

4.1 Dihedral Group

We first analyze the failure of our Navier-Stokes trained models to learn the dihedral group D_4 , a breakdown that presents across random seeds and mini-batches despite otherwise excellent pointwise accuracy and low untransformed test loss [see Figure 1]. Both architectures achieve competitive performance on the identity element e , 180 degree rotations r^2 , 90 degree rotations followed by a flip about the vertical axis sr , and 270 degree rotations followed by such a flip sr^3 , yet their response under the remaining group actions reveals a sharp and systematic lack of symmetry adherence. In this case, the relative SMSE increases catastrophically, by nearly 10^4 , indicating that the learned predictors are not even approximately equivariant when all group actions are considered. Such behavior is not entirely unexpected. The Navier-Stokes dataset carries a pronounced directional bias in feature space, inherited from the choice of initial conditions and the anisotropic, vorticity-dominated structure of the flows that is not invariant under reflections or 90 degree rotations. What is striking, however, is that the symmetry breaking is not merely a property of the forward map. The influence structure associated with these unlearned group actions is distinguished [see Figure 2]. Crucially, the group elements with catastrophic equivariance error are precisely those with suppressed cross-influence.

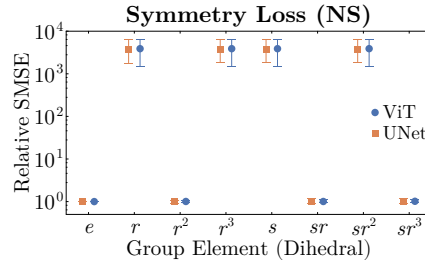


Figure 1: Dihedral group equivariance error on Navier-Stokes (NS) data. Relative SMSE evaluated on NS trajectories. Points denote medians over seeds and test examples; ranges indicate variability. Deviations from unity quantify dihedral symmetry breaking. For the analogous result involving CE data, see Figure 9.

168

Rather than providing a democratic sharing of coupling across the dihedral orbit, optimization drove our models into a region of the loss landscape that displays a clear demarcation between well-learned and challenging group elements, with the latter receiving subdominant influence. Notably, our UNets assign such challenging rotations near-zero cross-influence, even though these out-of-distribution (but physically permissible) transformations induce large gradients, indicating that training has converged to a basin with symmetry-incompatible local geometry. We emphasize that although the dihedral symmetry is physically valid and symmetry-transformed inputs generate large gradients, gradient signals do not accumulate coherently over symmetry operations. Instead, data-induced anisotropies are absorbed into the local loss geometry, inducing training dynamics that reinforce symmetry-breaking. Such rotations are in-distribution under reasonable deployment assumptions, meaning that

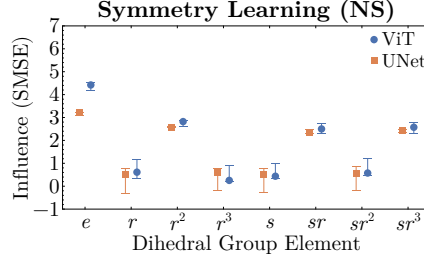


Figure 2: Dihedral group influence on NS data. Influence between an input and its square group rotated state on NS data. Points denote medians; ranges summarize inter-seed and inter-example variability, measuring coupling of learning dynamics along dihedral orbits. For the analogous result involving CE data, see Figure 10.

the observed decoupling reflects a zero-shot generalization gap between model performance on the training distribution and the operational distribution, indicating limited ability to generalize over the full data manifold. Our influence probe interrogates directions that are poorly aligned with the primary drift direction traced by the training flow. Hence, symmetry-related gradient signals enter only as subleading perturbations and fail to accumulate coherently, explaining how models achieve excellent pointwise accuracy on in-training-distribution data while converging to basins whose local geometry explicitly breaks dihedral symmetry.

Together, Figure 1 and Figure 2 resolve the apparent tension between low in-training-distribution test error and severe dihedral failure: under symmetry-agnostic training, the induced training dynamics do not propagate information across the entire D_4 orbit; consequently, neither model class achieves generalization over symmetry transformations, a necessary property for learning the underlying solution operator.

Our CE-trained models exhibit low square-group equivariance error [see Figure 9], accompanied by an influence profile that is distributed approximately uniformly across the D_4 orbit [see Figure 10]. This behavior is consistent with the symmetry operations being adequately represented within the training distribution, such that symmetry-agnostic optimization nevertheless induces orbit-wise coupling in the learned response geometry. This observation suggests that influence-based diagnostics can be used to reason about minimal data augmentation strategies.

4.2 Translation Group

For the translation group, we first evaluate purely horizontal and vertical translations across the full periodic spatial domain. Because the governing PDE applies uniformly in space, the learned dynamics should be equivariant under translations. In any given flow snapshot, nonlinear interactions occupy only a small number of localized regions, yet these structures may arise at arbitrary spatial locations. A model that learns translational equivariance will therefore treat such interactions consistently wherever they occur, a prerequisite for both capturing rare but dynamically significant events and enabling accurate long-time extrapolation.

Both architectures exhibit median relative equivariance error on the order of one percent or less, indicating that training largely confers translational symmetry generalizing capabilities. Notably, the relative equivariance error attained by our models [see Figure 3 and Figure 4] reflects genuine generalization over the translation group: we did not explicitly augment the training data with translated states, nor was either architecture designed to enforce exact equivariance over the entire translation group under consideration. In particular, translation invariance is violated by the distribution from which four quadrant initial conditions for both CE-RP and CE-RPUI flows were drawn. Nevertheless, we observe nontrivial cross-influence across translated states, indicating that training updates partially couple translation-related inputs, even without hard symmetry constraints. For this reason, we focus on the third quantile (Q3) when reporting translation equivariance error, which isolates cases with pronounced symmetry violation and exhibits a striking correspondence with the measured influence across translations. Focusing on the upper tail isolates examples for which translational symmetry is most weakly supported, making clear the relation between forward-pass failures and heterogeneity in the underlying gradient coupling. Indeed, the Q3 error profiles present differences

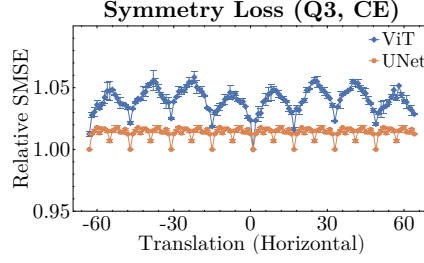


Figure 3: Horizontal-translation equivariance error on CE data. Third-quantile (Q3) relative SMSE as a function of horizontal translation. Markers denote medians across seeds; ranges summarize variability over seeds and test examples, emphasizing upper-tail symmetry breaking. For analogous results with NS data, see Figure 13.

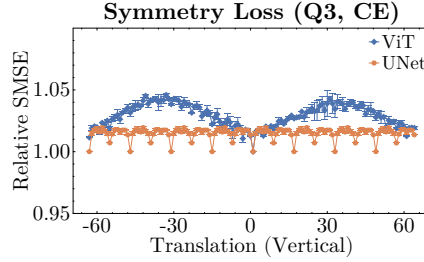


Figure 4: Vertical-translation equivariance error on CE data. Third-quantile relative SMSE under vertical translations. Markers denote medians across seeds; ranges summarize inter-seed and inter-example variability, isolating upper-tail error under symmetry perturbations. For analogous results with NS data, see Figure 12.

219 across architectures and anisotropic peculiarities that can be better understood when forward pass
 220 error is supplemented with influence function analysis [see Figure 5 and Figure 6].

221 For our ViTs, the translation response exhibits a pronounced axis-dependent resonance structure.
 222 Under horizontal (vertical) translations, the a periodicity of wavelength 16 (32) pixels indicates that
 223 the measured influence/equivariance observables couple anisotropically to the phase of the translation
 224 relative to the patch lattice, with different group elements emphasized in different directions. This
 225 axis-dependent anisotropy is consistent with a representational bias introduced by flattening the
 226 two-dimensional patch grid into a one-dimensional token sequence. While the patch embedding itself
 227 is convolutional of stride four, the subsequent permutation into tokens and channel mixing allows
 228 for a preferred ordering on spatial degrees of freedom, thereby breaking the manifest equivalence
 229 between horizontal and vertical directions, which impacts the learned response geometry.

230 In contrast, our UNets employ convolutional operators across contracting and expanding resolutions,
 231 with four down-sampling stages. The coarsest representation has spatial extent 8×8 , which promotes
 232 a comparatively smooth dependence of influence on translation, with only low-amplitude variation
 233 at shorter 8 pixel wavelengths and their subharmonics. Consistent with this architecture-level
 234 partial symmetry adherence, UNets exhibit similar influence profiles across horizontal and vertical
 235 translations.

236 In both cases, inspection of both forward-pass equivariance error and influence profile provides an
 237 explanatory mechanism for symmetry learning: resonant minima in the error profile coincide with
 238 structured extrema in the response, and vice-versa. Nevertheless, both models support cross-influence
 239 above the unit baseline, indicating that the learned response geometry supports nontrivial propagation
 240 across the translation orbit and thus generalization over translations, even in the absence of explicit
 241 translation augmentation or exact equivariance constraints.

242 Sporadic coupling over the group allows our ViTs to concentrate each learning signal on a subset of
 243 translation phases, producing stronger local responses at the cost of more heterogeneous orbit-wise
 244 coupling. Less rigid coupling between translation elements for the ViT may be advantageous when

paired with an overall larger peak response, as its updates are flexibly dispersed throughout the group.

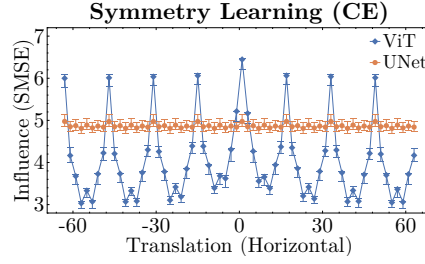


Figure 5: Horizontal-translation influence on CE data. Influence between an input and its horizontally translated state; structured dependence on translation distance reveals spatial coupling of learning dynamics. Markers denote medians; ranges indicate variability over seeds and test examples. For analogous results with NS data, see Figure 11.

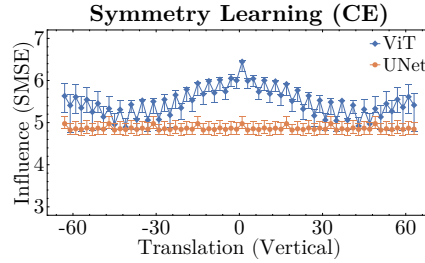


Figure 6: Vertical-translation influence on CE data. Same as the horizontal case, but for vertical shifts. Comparison with horizontal translations of Figure 5 reveals anisotropy in symmetry-coupled learning. Markers denote medians; ranges indicate variability over seeds and test examples. For analogous results with NS data, see Figure 14.

246

Beyond purely horizontal and vertical shifts, we also evaluate joint translations over a subset of the two-dimensional translation group. Figure 7 shows the resulting translation-equivariance error landscape for the ViT on CE data, while Figure 8 reports the corresponding influence landscape over the same grid. Together, these plots reveal that translation-induced symmetry breaking is organized into smooth, structured regions on the translation torus. Patterns in the forward-pass error align closely with features in the influence landscape: regions exhibiting elevated Q3 equivariance error coincide with translations for which parameter-update coupling is weak. This correspondence reinforces the interpretation that influence captures the effective resolution at which the model distinguishes translations. Given the periodic response exhibited by our models, we expect that this sub-sampling suffices to expose the geometry of translation generalization, without requiring exhaustive evaluation over the full $\mathbb{Z}_{128} \times \mathbb{Z}_{128}$ orbit. Indeed, the translation action is discrete and the influence curves are periodic, hence the full augmentation orbit $\mathbb{Z}_{128} \times \mathbb{Z}_{128}$ is massively redundant from the perspective of the learned response geometry. For data augmentation derived symmetry learning, our diagnostics suggest that exhaustive enumeration of over the full translation group may be unnecessary, one need only verify that unsampled translations lie in the same response-equivalence classes.

262 5 Related Work

Encoding symmetries as inductive biases has a long tradition in geometric deep learning. Group-equivariant CNNs generalize convolution to arbitrary groups with weight sharing across orbits Cohen and Welling (2016). Steerable CNNs make these ideas explicit by parameterizing kernels in group-steerable bases, including variants for volumetric data Cohen and Welling (2017); Weiler *et al.* (2018). For non-grid domains, tensor field networks Thomas *et al.* (2018) and E(n)-equivariant graph neural networks Garcia Satorras *et al.* (2021) extend manifest symmetry compliance to point clouds and

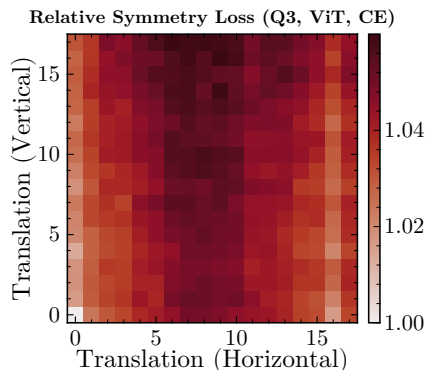


Figure 7: Translation-equivariance error landscape for ViT on CE data. Heatmap of third-quantile relative SMSE as a function of joint horizontal and vertical translations. Color encodes the magnitude of translation-induced symmetry breaking in the forward error. For analogous results with NS data, see Figure 20.

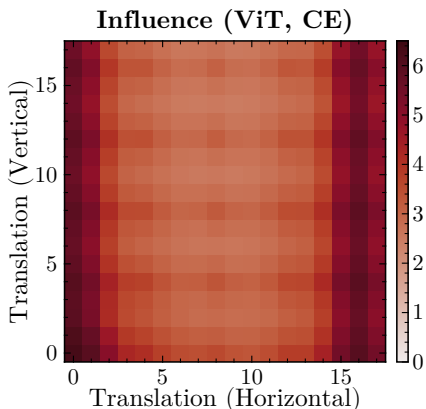


Figure 8: Translation-group influence landscape for ViT on CE data. Heatmap of influence between an input and its translated state over the translation grid. Color encodes the strength of parameter-update coupling across translated states. For analogous results with NS data, see Figure 19.

269 molecules. In practice, scientific data rarely obey exact symmetries; boundary conditions, material
 270 inhomogeneities, grid discretization, and measurement noise introduce systematic symmetry breaking.
 271 This motivates research into approximate Wang *et al.* (2022) and relaxed Finzi *et al.* (2021) attainment
 272 of equivariance. Furthermore, contemporary large-scale weather and climate models such as Aurora
 273 Bodnar *et al.* (2024) and ClimaX Nguyen *et al.* (2023) rely on data-driven learning of symmetry.
 274 Empirical and theoretical results substantiate this strategy: softening hard constraints can improve
 275 optimization and accuracy Wang *et al.* (2022). Relatedly, classic studies show that modern CNNs
 276 can be brittle to small shifts and rotations Azulay and Weiss (2019); Zhang (2019); Kayhan and van
 277 Gemert (2020).

278 In cases where a model lacking manifest symmetry is selected, probes are needed to both quantify and
 279 explain the degree of equivariance achieved after training. Several diagnostics test whether models
 280 learn symmetry, e.g., forward-pass checks to evaluate equivariance error under group actions Canez
 281 *et al.* (2024); Xie and Smidt (2025) and the Lie derivative metric, which quantifies infinitesimal
 282 equivariance with layerwise decomposition Gruver *et al.* (2022). Yet, spot tests with these metrics are
 283 insufficient to explain the mechanism underlying symmetry learning. Such diagnostics characterize
 284 equivariance of the learned map, but do not reveal whether training dynamics actually propagate
 285 information across symmetry-related states. To better understand the degree to which a given model
 286 exhibits equivariance, one can relate observed behavior to architectural choices and training dynamics.

A central approach in explainable AI is to interrogate the loss and its derivatives to connect predictive behavior with training signals. Related work on gradient geometry links cross-example parameter update structure to out-of-sample performance: stiffness and coherent gradients capture alignment, and local elasticity studies stability under SGD updates on distant samples Fort *et al.* (2020); Chatterjee (2020); He and Su (2020). Euclidean influence functions trace predictions to training data but are delicate in deep, non-convex regimes, motivating curvature-aware variants Koh and Liang (2020); Basu *et al.* (2020); Jacot *et al.* (2020); Fort and Ganguli (2019).

To our knowledge, this work is the first to analyze symmetry learning through the lens of influence functions, measuring how parameter updates induced by an input propagate to its symmetry-transformed counterparts. Unlike prior diagnostics that evaluate equivariance at the level of the learned map, our approach probes whether training dynamics themselves propagate information across symmetry orbits.

For a discussion of limitations to our work, see Appendix F. The code used in this work is available at <https://anonymous.4open.science/r/PDEHats-B7BF/>.

6 Conclusion

Our analysis sharpens recent insights on the interplay between inductive bias, optimization, and symmetry learning Canez *et al.* (2024). We show that test time equivariance error is determined by how training dynamics distribute influence across group orbits. While symmetry-agnostic architectures can attain low test error, such models allow for disparate allocation of influence across orbits, producing solutions that interpolate accurately yet fail to learn the equivariance structure of the underlying solution operator.

On two PDE emulation tasks, our ViTs converged rapidly, but did so with freely specialized gradients, leading to strong on-diagonal responses that sacrifice faithful symmetry representation. In this regime, low test error coexists with idiosyncratic orbit-wise coupling, signaling convergence to symmetry-incompatible basins: the result is high predictive accuracy but limited physical consistency, characteristic of powerful interpolators rather than genuine physics emulators; apparent accuracy in the absence of gradient coherence is an indicator of reduced robustness under symmetry transformations. By contrast, manifestly equivariant group convolution layers, such as those in UNets, promote data efficiency and principled generalization through uniform gradient coupling across symmetry orbits, but this rigidity can slow convergence and hinder optimization. Even when typical equivariance errors are relatively low, we find that model failure to respect symmetry on select challenging examples is explainable by a lack of coherent orbit-wise gradient sharing, indicating a failure of the symmetry learning mechanism to apply wholly across the dataset.

By explicitly measuring cross-influence between symmetry-related states, our framework reveals whether learning dynamics propagate information coherently along an orbit or instead distribute influence in a way that precludes convergence to generalizing solutions. This provides a novel diagnostic for distinguishing models that learn to exploit shared structure from those that effectively assemble collections of local estimators. Tying symmetry generalization directly to the geometry of the learned loss landscape complements standard equivariance accuracy metrics and provides a clearer criterion for when apparent performance reflects genuine physics learning. Such diagnostics are essential for building trust in scientific machine learning systems, particularly in applications requiring robustness under symmetry transformations. Looking forward, this perspective motivates the development of approximate or relaxed symmetry mechanisms that retain sufficient structure to guide generalization while preserving the flexibility needed for efficient optimization, potentially reconciling the scalability of transformers with the principled foundations of equivariant modeling.

References

- J. Brandstetter, D. Worrall, and M. Welling, Message passing neural pde solvers (2023), arXiv:2202.03376 [cs.LG].
- M. Herde, B. Raonić, T. Rohner, R. Käppeli, R. Molinaro, E. de Bézenac, and S. Mishra, Poseidon: Efficient foundation models for pdes (2024), arXiv:2405.19101 [cs.LG].

337 M. Takamoto, T. Praditia, R. Leiteritz, D. MacKinlay, F. Alesiani, D. Pflüger, and M. Niepert,
338 Pdebench: An extensive benchmark for scientific machine learning (2024), arXiv:2210.07182
339 [cs.LG] .

340 P. Lippe, B. S. Veeling, P. Perdikaris, R. E. Turner, and J. Brandstetter, Pde-refiner: Achieving
341 accurate long rollouts with neural pde solvers (2023), arXiv:2308.05732 [cs.LG] .

342 J. K. Gupta and J. Brandstetter, Towards multi-spatiotemporal-scale generalized pde modeling (2022),
343 arXiv:2209.15616 [cs.LG] .

344 R. Ohana, M. McCabe, L. Meyer, R. Morel, F. J. Agocs, M. Beneitez, M. Berger, B. Burkhart,
345 K. Burns, S. B. Dalziel, D. B. Fielding, D. Fortunato, J. A. Goldberg, K. Hirashima, Y.-F. Jiang,
346 R. R. Kerswell, S. Maddu, J. Miller, P. Mukhopadhyay, S. S. Nixon, J. Shen, R. Watteaux, B. R.-S.
347 Blancard, F. Rozet, L. H. Parker, M. Cranmer, and S. Ho, The well: a large-scale collection of
348 diverse physics simulations for machine learning (2025), arXiv:2412.00568 [cs.LG] .

349 T. Akhound-Sadegh, L. Perreault-Levasseur, J. Brandstetter, M. Welling, and S. Ravanbakhsh, Lie
350 point symmetry and physics informed networks (2023), arXiv:2311.04293 [cs.LG] .

351 W. G. Gregory, D. W. Hogg, B. Blum-Smith, M. T. Arias, K. W. K. Wong, and S. Villar, Equivariant
352 geometric convolutions for emulation of dynamical systems (2024), arXiv:2305.12585 [cs.LG] .

353 N. Gruver, M. Finzi, M. Goldblum, and A. G. Wilson, arXiv:2210.02984 (2022).

354 S. Fort, P. K. Nowak, S. Jastrzebski, and S. Narayanan, Stiffness: A new perspective on generalization
355 in neural networks (2020), arXiv:1901.09491 [cs.LG] .

356 B. Zhao, R. M. Gower, R. Walters, and R. Yu, Improving convergence and generalization using
357 parameter symmetries (2024), arXiv:2305.13404 [cs.LG] .

358 J. Brandstetter, M. Welling, and D. E. Worrall, in *Proceedings of the 39th International Conference on*
359 *Machine Learning*, Proceedings of Machine Learning Research, Vol. 162, edited by K. Chaudhuri,
360 S. Jegelka, L. Song, C. Szepesvari, G. Niu, and S. Sabato (PMLR, 2022) pp. 2241–2256.

361 D. Arpit, S. Jastrzebski, N. Ballas, D. Krueger, E. Bengio, M. S. Kanwal, T. Maharaj, A. Fischer,
362 A. Courville, Y. Bengio, and S. Lacoste-Julien, A closer look at memorization in deep networks
363 (2017), arXiv:1706.05394 [stat.ML] .

364 H. He and W. J. Su, The local elasticity of neural networks (2020), arXiv:1910.06943 [cs.LG] .

365 S. Chatterjee, Coherent gradients: An approach to understanding generalization in gradient descent-
366 based optimization (2020), arXiv:2002.10657 [cs.LG] .

367 Anonymous, Generalization vs. memorization in autoregressive deep learning: Or, examining tempo-
368 ral decay of gradient coherence (2026).

369 A. Pal, Lux: Explicit Parameterization of Deep Neural Networks in Julia (2023a).

370 A. Pal, On Efficient Training & Inference of Neural Differential Equations (2023b).

371 M. Innes, CoRR **abs/1810.07951** (2018), arXiv:1810.07951 .

372 A. Jacot, F. Gabriel, and C. Hongler, Neural tangent kernel: Convergence and generalization in neural
373 networks (2020), arXiv:1806.07572 [cs.LG] .

374 P.-A. Absil, R. Mahony, and R. Sepulchre, *Optimization Algorithms on Matrix Manifolds* (Princeton
375 University Press, Princeton, NJ, 2008).

376 S. Fort and S. Ganguli, Emergent properties of the local geometry of neural loss landscapes (2019),
377 arXiv:1910.05929 [cs.LG] .

378 J. Martens, Journal of Machine Learning Research **21**, 1 (2020).

379 J. Lu, D. Ko, and T. Chang, Journal of the American Statistical Association **92**, 1572 (1997).

380 S. Héritier and E. Ronchetti, Journal of the American Statistical Association **89**, 897 (1994).

381 R. Zhang, in *Proceedings of the 36th International Conference on Machine Learning*, Proceedings of
382 Machine Learning Research, Vol. 97, edited by K. Chaudhuri and R. Salakhutdinov (PMLR, 2019)
383 pp. 7324–7334.

384 A. Azulay and Y. Weiss, *Journal of Machine Learning Research* **20**, 1 (2019).

385 O. S. Kayhan and J. C. van Gemert, On translation invariance in cnns: Convolutional layers can
386 exploit absolute spatial location (2020), arXiv:2003.07064 [cs.CV] .

387 S. Wang, A. K. Bhartari, B. Li, and P. Perdikaris, Gradient alignment in physics-informed neural
388 networks: A second-order optimization perspective (2025), arXiv:2502.00604 [cs.LG] .

389 R. Wang, R. Walters, and R. Yu, Approximately equivariant networks for imperfectly symmetric
390 dynamics (2022), arXiv:2201.11969 [cs.LG] .

391 T. S. Cohen and M. Welling, in *Proceedings of the 33rd International Conference on Machine*
392 *Learning*, PMLR, Vol. 48 (2016) pp. 2990–2999.

393 T. S. Cohen and M. Welling, in *ICLR* (2017).

394 M. Weiler, M. Geiger, M. Welling, W. Boomsma, and T. Cohen, arXiv:1807.02547 (2018).

395 N. Thomas, T. Smidt, S. Kearnes, L. Yang, L. Li, K. Kohlhoff, and P. Riley, arXiv:1802.08219
396 (2018).

397 V. Garcia Satorras, E. Hoogeboom, and M. Welling, in *Proceedings of the 38th International*
398 *Conference on Machine Learning*, PMLR, Vol. 139 (2021) pp. 9323–9332.

399 M. Finzi, G. Benton, and A. G. Wilson, Residual pathway priors for soft equivariance constraints
400 (2021), arXiv:2112.01388 [cs.LG] .

401 C. Bodnar, W. P. Bruinsma, A. Lucic, M. Stanley, A. Vaughan, J. Brandstetter, P. Garvan, M. Riechert,
402 J. A. Weyn, H. Dong, J. K. Gupta, K. Thambiratnam, A. T. Archibald, C.-C. Wu, E. Heider,
403 M. Welling, R. E. Turner, and P. Perdikaris, A foundation model for the earth system (2024),
404 arXiv:2405.13063 [physics.ao-ph] .

405 T. Nguyen, J. Brandstetter, A. Kapoor, J. K. Gupta, and A. Grover, Climax: A foundation model for
406 weather and climate (2023), arXiv:2301.10343 [cs.LG] .

407 D. Canez, N. Midavaine, T. Stessen, J. Fan, S. Arias, and A. Garcia, Effect of equivariance on training
408 dynamics (2024).

409 Y. Xie and T. Smidt, A tale of two symmetries: Exploring the loss landscape of equivariant models
410 (2025), arXiv:2506.02269 [cs.LG] .

411 P. W. Koh and P. Liang, Understanding black-box predictions via influence functions (2020),
412 arXiv:1703.04730 [stat.ML] .

413 S. Basu, X. You, and S. Feizi, in *Proceedings of the 37th International Conference on Machine*
414 *Learning*, PMLR, Vol. 119 (2020) pp. 715–724.

415 Z. Ye, X. Huang, L. Chen, H. Liu, Z. Wang, and B. Dong, in *ICLR 2024 Workshop on*
416 *AI4DifferentialEquations In Science* (2024).

417 Z. Liu, Y. Lin, Y. Cao, H. Hu, Y. Wei, Z. Zhang, S. Lin, and B. Guo, Swin transformer: Hierarchical
418 vision transformer using shifted windows (2021), arXiv:2103.14030 [cs.CV] .

419 Z. Li, N. Kovachki, K. Azizzadenesheli, B. Liu, K. Bhattacharya, A. Stuart, and A. Anandkumar,
420 Fourier neural operator for parametric partial differential equations (2021), arXiv:2010.08895
421 [cs.LG] .

422 S. Subramanian, P. Harrington, K. Keutzer, W. Bhimji, D. Morozov, M. Mahoney, and A. Gholami,
423 Towards foundation models for scientific machine learning: Characterizing scaling and transfer
424 behavior (2023), arXiv:2306.00258 [cs.LG] .

425 L. Lu, P. Jin, G. Pang, Z. Zhang, and G. E. Karniadakis, *Nature Machine Intelligence* **3**, 218–229
426 (2021).

427 D. Orban and M. Arioli, *Iterative Solution of Symmetric Quasi-Definite Linear Systems* (Society for
428 Industrial and Applied Mathematics, Philadelphia, PA, 2017).

429 A. Montoison and D. Orban, *Journal of Open Source Software* **8**, 5187 (2023).

430 T. George, NNGeometry: Easy and Fast Fisher Information Matrices and Neural Tangent Kernels in
431 PyTorch (2021).

432 TransferLab, pyDVL (2024).

433 D. P. Kingma and J. Ba, Adam: A method for stochastic optimization (2017), arXiv:1412.6980
434 [cs.LG] .

435 A Data Selection.

436 PDEGym data is particularly valuable for studying the progression from a linear wave regime with
 437 discontinuities to fully developed turbulence, a crossover that poses computational and analytical
 438 challenges due to the presence of sharp wave-fronts and emergent nonlinear interactions. While the
 439 CE flows exhibit comparable large-scale structures, they also display qualitatively distinct behaviors.
 440 In particular, CE-RPUI initial configurations give rise to complex finger-like instabilities in the
 441 flow field that are absent or less pronounced in both CE-RP and CE-CRP. In total, we used 6, 500
 442 trajectories for each of the three classes of initial conditions; for each trajectory, we used the first 16
 443 time steps, for total of approximately 110, 000 training pairs requiring greater than 150 GB memory.

444 B Model Selection.

445 We focus on UNet and ViT backbones because they represent the dominant scalable image-to-image
 446 paradigms currently used in neural PDE emulation and PDE foundation modeling. Convolutional
 447 backbones derived from UNets remain standard architectures for grid-based surrogate modeling and
 448 large-scale benchmarking efforts Takamoto *et al.* (2024); Gupta and Brandstetter (2022). Likewise,
 449 transformer-based backbones now underpin many recent PDE foundation models and large-scale
 450 operator learners, including Poseidon and PDEformer Herde *et al.* (2024); Ye *et al.* (2024), in
 451 addition to several contemporary multi-physics ? and shifting window transformer Liu *et al.* (2021)
 452 architectures. Fourier neural operators provide an important complementary class of operator-
 453 learning models Li *et al.* (2021); Subramanian *et al.* (2023); however, because their representations
 454 are constructed from truncated Fourier modes, the resulting plane-wave basis is less naturally suited
 455 to the spatially localized discontinuities and sharp shock fronts characteristic of compressible flow
 456 evolution Ohana *et al.* (2025). Consistent with this expectation, we found FNO variants to be
 457 noncompetitive with both UNet and ViT backbones on the flows considered in this work. DeepONets
 458 provide a foundational operator-learning formalism Lu *et al.* (2021), but are not the prevailing
 459 architecture for contemporary large-scale PDE solution operator emulation.

460 C Loss Sensitivity as Gradient Overlap

461 The influence diagnostic admits a simple interpretation as a first-order sensitivity of the loss function.
 462 Let C_p denote the cost associated with a perturbing example p , and let the induced parameter
 463 displacement be $\delta\theta_p = -\eta^{-1}\nabla_{\theta}C_p$, where η is the Gauss-Newton metric. The corresponding first-
 464 order change in the cost of a response example r is $C_r(\theta + \delta\theta_p) - C_r(\theta) \approx -\langle \nabla_{\theta}C_r, \eta^{-1}\nabla_{\theta}C_p \rangle$.
 465 Thus, gradient alignment measures the first-order sensitivity of example r to an update generated by
 466 example p . Positive alignment indicates that the perturbing update acts coherently on the response
 467 example, whereas weak or negative alignment indicates that the corresponding learning signals are
 468 decoupled or conflicting.

469 D Determination of η^{-1}

470 Influence-function computations in deep learning are commonly performed either through iterative
 471 matrix-free solvers, such as conjugate-gradient methods Orban and Arioli (2017); Montoison and Or-
 472 ban (2023), or through scalable approximations to the curvature operator George (2021); TransferLab
 473 (2024). While such approaches are often necessary in high-dimensional parameter spaces, iterative
 474 methods require repeated Hessian-vector products and convergence tolerances, whereas low-rank or
 475 diagonal approximations generically do not provide controlled error estimates.

476 In the present setting, however, the influence observable depends only on metric-weighted overlaps
 477 between example gradients. Let $G_{\alpha}^n = \partial_{\alpha}C^n$ denote the gradient associated with mini-batch example
 478 n . Performing a QR decomposition on G yields an orthonormal basis Q that projects onto the
 479 gradient subspace. Hence, $\Pi^{nm} = J_{\mu}^n Q_{\alpha}^{\mu} \hat{\eta}^{ab} Q_b^{\nu} J_{\nu}^m$, where, $\hat{\eta}_{ab} = Q_a^{\nu} \eta_{\mu\nu} Q_b^{\mu}$ can be efficiently
 480 inverted. Since the dimension of the gradient-spanned subspace is bounded by the mini-batch
 481 size rather than the total number of trainable parameters, the resulting inversion is both exact and

482 computationally efficient, eliminating the need for uncontrolled curvature approximations or iterative
483 parameter-space solves.

484 For CE and NS data, matrix elements of Π were determined for 18 different mini-batches, each of
485 which contained 3 trajectories corresponding to distinct initial conditions over 16 time-steps, for each
486 seed of each model architecture trained. In practice, we additionally include an ℓ_2 -type regularizer
487 corresponding to the weight-decay term used during AdamW optimization Kingma and Ba (2017),
488 $\eta_{\mu\nu} \rightarrow \eta_{\mu\nu} + \lambda \delta_{\mu\nu}$, with λ weakly lifting the zero modes of η .

489 **E Influence Normalization and Interpretation**

490 As a sensitivity measure, the influence function possesses no canonical scale. Common choices
491 include normalization by gradient magnitude, yielding the cosine similarity, normalization by trace
492 or Frobenius norms, or alternatively leaving the response unnormalized altogether. Taking inspiration
493 from the standardized influence function literature Héritier and Ronchetti (1994); Lu *et al.* (1997),
494 we report influence normalized with respect to the empirical variance of perturbation sensitivities
495 within each mini-batch, such that the baseline corresponding to unstructured stochastic variability
496 naturally defines the reference scale of unity. Under this normalization, departures from one directly
497 quantify influence beyond that expected from random mini-batch fluctuations, providing a principled
498 scale for interpreting both amplified self-responses and suppressed cross-responses.

499 **F Limitations**

500 Certain physically relevant symmetries remain outside the scope of the present study. Galilean
501 boosts, in particular, impose a highly restrictive constraint: exact equivariance can be achieved only
502 by affine transformations, while generic nonlinear architectures can at best satisfy this symmetry
503 approximately Wang *et al.* (2022). It would be interesting to extend our influence-based analysis
504 to probe the response under small Galilean boosts. We did not examine the scaling symmetry of
505 the continuum Navier-Stokes equations. In practical settings this symmetry is generically broken
506 by spatial discretization, numerical regularization, and coarse graining of the data, rendering its
507 faithful assessment ambiguous within the present framework. Finally, our empirical analysis is
508 limited to UNet and ViT architectures, chosen as representative backbones of foundation models
509 widely used in scientific machine learning. For architectures that enforce exact equivariance by
510 construction, influence across symmetry orbits would be uniform by design. While such architectures
511 guarantee symmetry by construction, they do not address whether symmetry can be learned implicitly
512 under realistic data and optimization constraints, which is the regime targeted by modern large-
513 scale scientific models. Our focus is therefore on settings in which symmetry learning is implicit
514 and contingent on training dynamics rather than guaranteed a priori. Furthermore, our diagnostics
515 characterize symmetry learning relative to the training distribution; when symmetry transformations
516 are poorly represented in data, influence-based measures reflect this bias rather than an intrinsic
517 architectural limitation.

518 Our analysis is restricted to two widely used architectural backbones, namely UNets and Vision
519 Transformers. While these models represent dominant paradigms in contemporary large-scale PDE
520 emulation, they do not exhaust the space of operator-learning methods. Consequently, the degree to
521 which our measured gradient coherence properties depend on the chosen backbone rather than on the
522 autoregressive training paradigm itself remains only partially resolved.

523 **G Supplementary Figures**

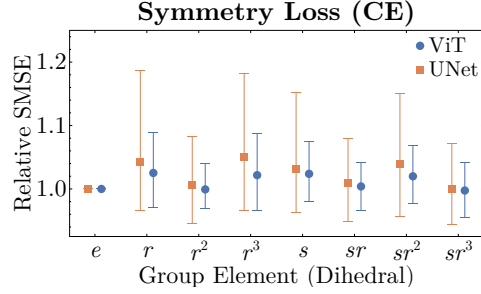


Figure 9: Dihedral (square-group D_4) equivariance error on CE data. For each group element $g \in D_4$ (quarter-rotations r and a reflection s), we report the relative SMSE, $\text{SMSE}(g \cdot x)/\text{SMSE}(x)$, at fixed time and within class. Markers denote medians over test examples and random seeds; vertical ranges indicate inter-seed and inter-example variability. Values exceeding unity indicate symmetry breaking in the model predictions.

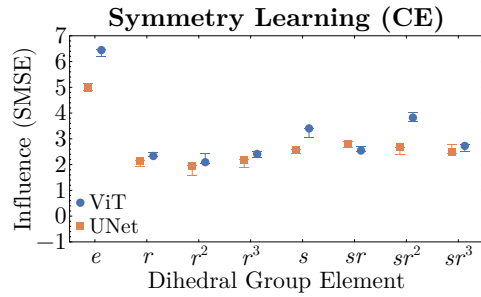


Figure 10: Dihedral group influence on CE data. Influence between an input and its square group transformed counterpart. Influence measures gradient overlap in the NTK-geometry, indicating coupling of parameter updates along the dihedral orbit. Markers denote medians; ranges indicate variability over seeds and test examples.

NeurIPS Paper Checklist

1. Claims

Question: Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?

Answer: [Yes]

Justification: The abstract and introduction accurately reflect the paper’s central contributions: a geometry-aware influence-function framework for probing symmetry learning in neural PDE emulators, an exact and efficient hat-matrix weighted gradient-alignment diagnostic computed in the mini-batch gradient subspace, and an empirical analysis relating equivariance error to orbit-wise gradient coherence across dihedral and translation symmetry groups. The manuscript further demonstrates how local loss-landscape geometry governs cross-orbit information propagation and symmetry-consistent generalization in UNet and Vision Transformer architectures trained on compressible Euler and Navier–Stokes systems. These claims are substantiated by both the theoretical development and the experimental results presented throughout the paper.

2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: [Yes]

Justification: The paper discusses several limitations. In particular, the analysis is restricted to autoregressive UNet and ViT architectures trained on PDEGym-derived compressible Euler and Navier–Stokes datasets. The manuscript also discusses limitations associated with restricted symmetry classes and dependence on the training distribution.

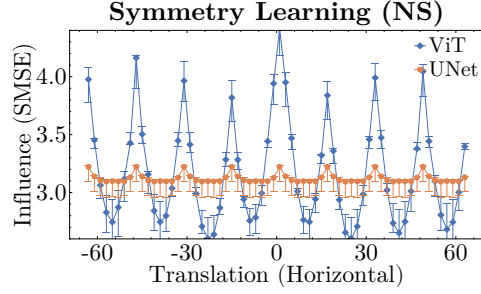


Figure 11: Horizontal-translation influence on NS data. Influence between an input and its horizontal translation. Points denote medians; ranges indicate variability over seeds and test examples.

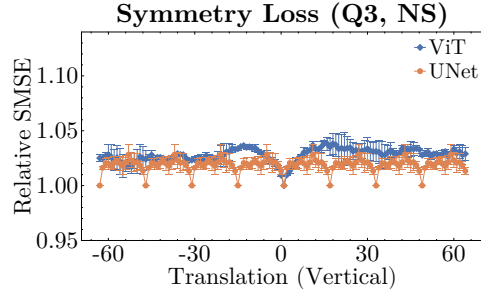


Figure 12: Vertical-translation equivariance error on NS data (third quartile). Third-quantile relative SMSE as a function of vertical translation. Points denote medians; ranges highlight upper-tail symmetry breaking.

- 546 3. Theory assumptions and proofs
- 547 Question: For each theoretical result, does the paper provide the full set of assumptions and
- 548 a complete (and correct) proof?
- 549 Answer: [No]
- 550 Justification: The manuscript develops a formal response-theoretic framework for influence
- 551 functions and proximal geometry, but does not present fully rigorous theorem-proof style
- 552 derivations for all statements. Instead, the work emphasizes physically motivated derivations,
- 553 definitions, and constructions supported by references to established literature.
- 554 4. Experimental result reproducibility
- 555 Question: Does the paper fully disclose all the information needed to reproduce the main ex-
- 556 perimental results of the paper to the extent that it affects the main claims and/or conclusions
- 557 of the paper (regardless of whether the code and data are provided or not)?
- 558 Answer: [Yes]

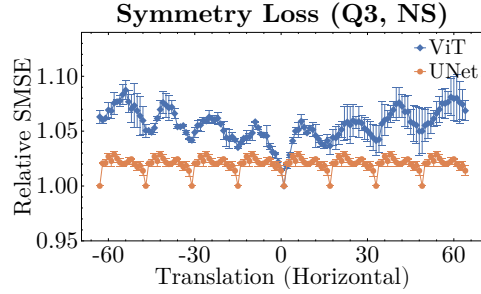


Figure 13: Horizontal-translation equivariance error on NS data (third quartile). Third-quantile relative SMSE under horizontal translations. Points denote medians across seeds; ranges emphasize upper-tail translation-induced error.

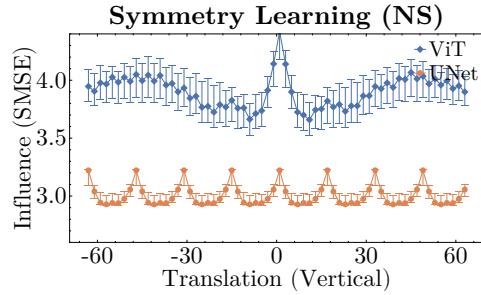


Figure 14: Vertical-translation influence on NS data. Influence between an input and its vertical translation. Points denote medians; ranges summarize variability across seeds and test examples.

559 Justification: The paper specifies the datasets, architectures, parameter counts, optimizer,
 560 learning rates, regularization, batch sizes, hardware configuration, number of seeds, training
 561 setup, and influence-function computation procedure. Details concerning the QR-based low-
 562 rank inversion strategy and observational definitions are also provided, enabling reproduction
 563 of the main experiments and conclusions.

564 5. Open access to data and code

565 Question: Does the paper provide open access to the data and code, with sufficient instruc-
 566 tions to faithfully reproduce the main experimental results, as described in supplemental
 567 material?

568 Answer: [Yes]

569 Justification: The experiments rely on publicly available datasets, namely PDEGym. The
 570 implementation code and training scripts are released at submission time. The manuscript

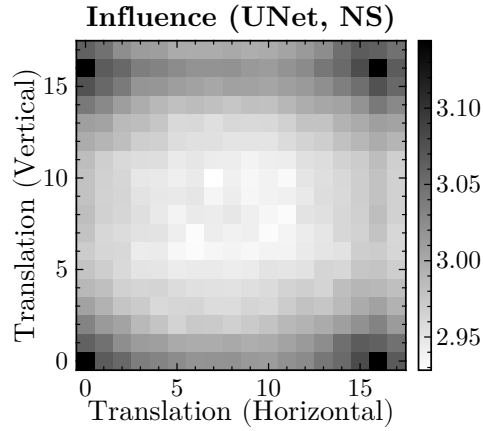


Figure 15: Translation-group influence landscape for UNet on NS data. Heatmap of Influence across joint translations quantifying how UNet parameter updates couple translated NS states.

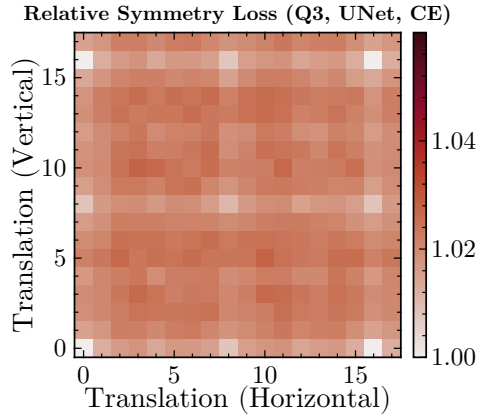


Figure 16: Translation-equivariance error landscape for UNet on CE data. Heatmap of third-quantile relative SMSE versus joint translations. Smoother structure relative to ViT reflects convolutional inductive bias modulated by multiscale pooling and upsampling. For analogous results with NS data, see Figure 17.

571 provides detailed methodological and experimental descriptions sufficient for independent
572 reproduction.

573 6. Experimental setting/details

574 Question: Does the paper specify all the training and test details (e.g., data splits, hyperpa-
575 rameters, how they were chosen, type of optimizer) necessary to understand the results?

576 Answer: [Yes]

577 Justification: The paper provides the dataset composition, number of trajectories, spatial res-
578 olution, train/test protocol, optimizer choice, learning rates, weight decay, mini-batch sizes,
579 architecture details, parameter counts, checkpoint structure, and hardware configuration.
580 The influence observables and normalization conventions are also defined explicitly.

581 7. Experiment statistical significance

582 Question: Does the paper report error bars suitably and correctly defined or other appropriate
583 information about the statistical significance of the experiments?

584 Answer: [Yes]

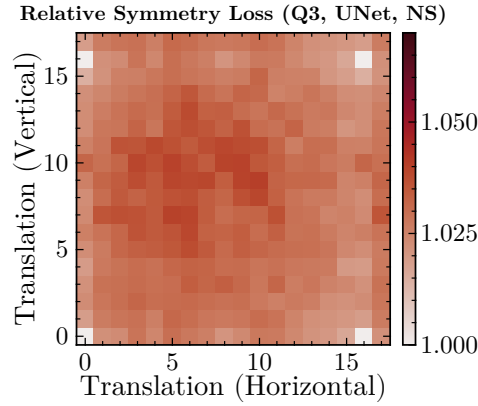


Figure 17: Translation-equivariance error landscape for UNet on NS data. Heatmap of third-quantile relative SMSE, providing a two-dimensional view of residual translation symmetry loss.

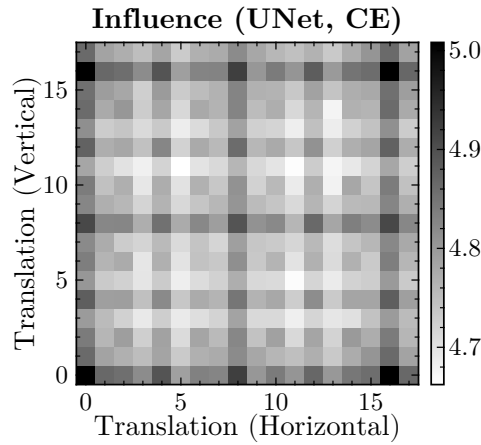


Figure 18: Translation-group influence landscape for UNet on CE data. Heatmap of influence across joint translations. Spatial structure diagnoses how UNet training updates propagate along translation orbits. For analogous results with NS data, see Figure 15.

585 Justification: Figures throughout the paper include range bars representing variability across
 586 multiple training seeds and mini-batch samplings. The manuscript explicitly states that
 587 multiple independent training runs were performed and explains the interpretation of the
 588 reported variability measures.

589 8. Experiments compute resources

590 Question: For each experiment, does the paper provide sufficient information on the com-
 591 puter resources (type of compute workers, memory, time of execution) needed to reproduce
 592 the experiments?

593 Answer: [Yes]

594 Justification: The paper specifies that training was performed in distributed mode across two
 595 40GB NVIDIA A100 GPUs.

596 9. Code of ethics

597 Question: Does the research conducted in the paper conform, in every respect, with the
 598 NeurIPS Code of Ethics <https://neurips.cc/public/EthicsGuidelines?>

599 Answer: [Yes]

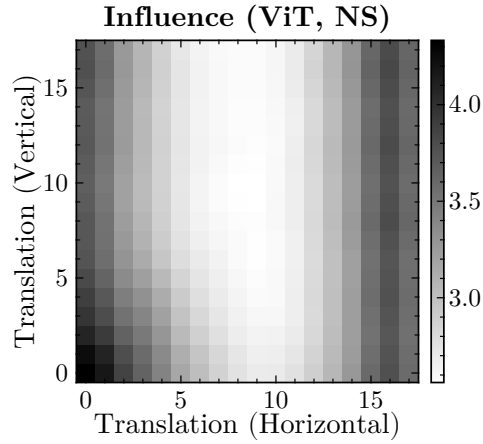


Figure 19: Translation-group influence landscape for ViT on NS data. Heatmap of Influence between an example and its translated state . Color encodes the strength of translation-orbit coupling in ViT learning dynamics.

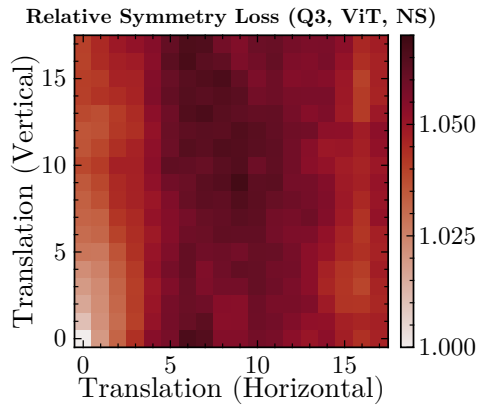


Figure 20: Translation-equivariance error landscape for ViT on NS data. Heatmap of third-quantile relative SMSE over joint translations, identifying translation pairs that induce elevated upper-tail error.

600 Justification: The work studies neural PDE emulators trained on data that may involve human sub-
 601 jects, private data, surveillance, or sensitive information. The research was conducted in
 602 accordance with the NeurIPS Code of Ethics.

603 10. Broader impacts

604 Question: Does the paper discuss both potential positive societal impacts and negative
 605 societal impacts of the work performed?

606 Answer: [Yes]

607 Justification: The paper discusses how improved diagnostics for identifying memorization
 608 and generalization in scientific machine learning could improve the reliability of surrogate
 609 models used in scientific computing. It also notes that unreliable surrogate models may
 610 produce misleading or physically inconsistent predictions if deployed without sufficient
 611 safeguards or diagnostic validation.

612 11. Safeguards

613 Question: Does the paper describe safeguards that have been put in place for responsible
614 release of data or models that have a high risk for misuse (e.g., pre-trained language models,
615 image generators, or scraped datasets)?

616 Answer: [N/A]

617 Justification: The paper does not involve high-risk generative models, scraped datasets, or
618 dual-use foundation models. The datasets used are publicly available scientific simulation
619 benchmarks from the existing literature.

620 12. Licenses for existing assets

621 Question: Are the creators or original owners of assets (e.g., code, data, models), used in
622 the paper, properly credited and are the license and terms of use explicitly mentioned and
623 properly respected?

624 Answer: [Yes]

625 Justification: The paper properly cites PDEGym and all referenced model architectures and
626 prior methods. Existing datasets and frameworks are credited throughout the main text and
627 references.

628 13. New assets

629 Question: Are new assets introduced in the paper well documented and is the documentation
630 provided alongside the assets?

631 Answer: [No]

632 Justification: The paper does not release a new dataset, benchmark, or model artifact at
633 submission time. The primary contribution is methodological and diagnostic rather than the
634 release of a new public asset.

635 14. Crowdsourcing and research with human subjects

636 Question: For crowdsourcing experiments and research with human subjects, does the paper
637 include the full text of instructions given to participants and screenshots, if applicable, as
638 well as details about compensation (if any)?

639 Answer: [N/A]

640 Justification: The paper does not involve crowdsourcing or research with human subjects.

641 15. Institutional review board (IRB) approvals or equivalent for research with human subjects

642 Question: Does the paper describe potential risks incurred by study participants, whether
643 such risks were disclosed to the subjects, and whether Institutional Review Board (IRB)
644 approvals (or an equivalent approval/review based on the requirements of your country or
645 institution) were obtained?

646 Answer: [N/A]

647 Justification: The paper does not involve human subjects, participant studies, or crowdsourc-
648 ing experiments.

649 16. Declaration of LLM usage

650 Question: Does the paper describe the usage of LLMs if it is an important, original, or
651 non-standard component of the core methods in this research? Note that if the LLM is used
652 only for writing, editing, or formatting purposes and does not impact the core methodology,
653 scientific rigor, or originality of the research, declaration is not required.

654 Answer: [N/A]

655 Justification: LLMs are not part of the scientific methodology, experimental framework, or
656 core technical contribution of this work.